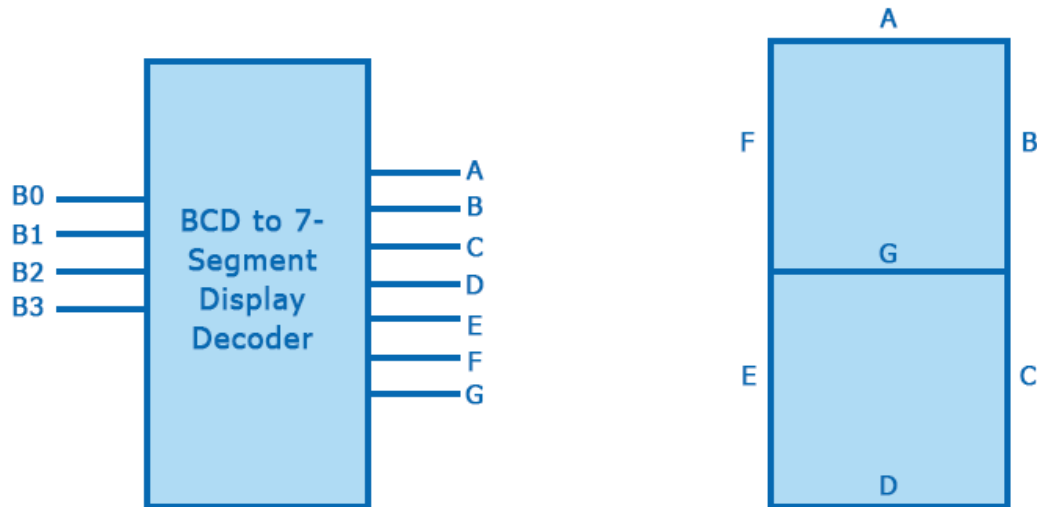


Day 21

BCD to 7-Segment Display Decoder in Verilog

Circuit Diagram :



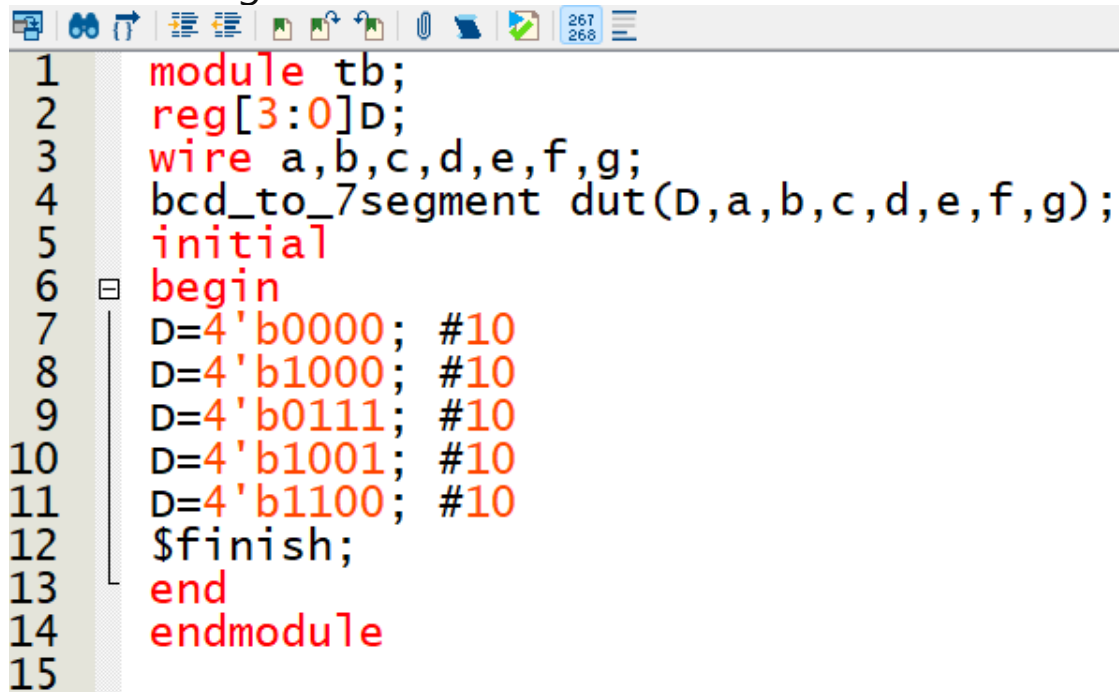
Verilog Code

BCD to 7-Segment Verilog code

```
1 module bcd_to_7segment(D,a,b,c,d,e,f,g );
2 input [3:0]D;
3 output reg a,b,c,d,e,f,g;
4 always @(*)
5 begin
6 case(D)
7 4'b0000: {a,b,c,d,e,f,g} = 7'b1111110;
8 4'b0001: {a,b,c,d,e,f,g} = 7'b0110000;
9 4'b0010: {a,b,c,d,e,f,g} = 7'b1101101;
10 4'b0011: {a,b,c,d,e,f,g} = 7'b1111001;
11 4'b0100: {a,b,c,d,e,f,g} = 7'b0110011;
12 4'b0101: {a,b,c,d,e,f,g} = 7'b1011011;
13 4'b0110: {a,b,c,d,e,f,g} = 7'b1011111;
14 4'b0111: {a,b,c,d,e,f,g} = 7'b1110000;
15 4'b1000: {a,b,c,d,e,f,g} = 7'b1111111;
16 4'b1001: {a,b,c,d,e,f,g} = 7'b1111011;
17 default: {a,b,c,d,e,f,g} = 7'b0000000;
18 endcase
19 end
20 endmodule
```

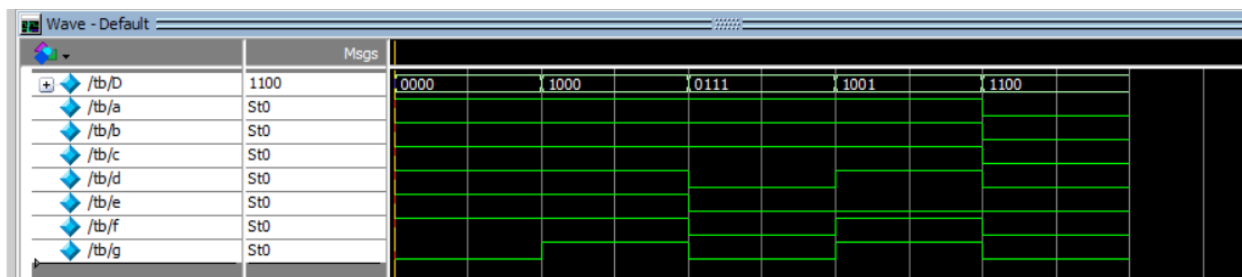
Testbench Code

BCD to 7-Segment Tb code

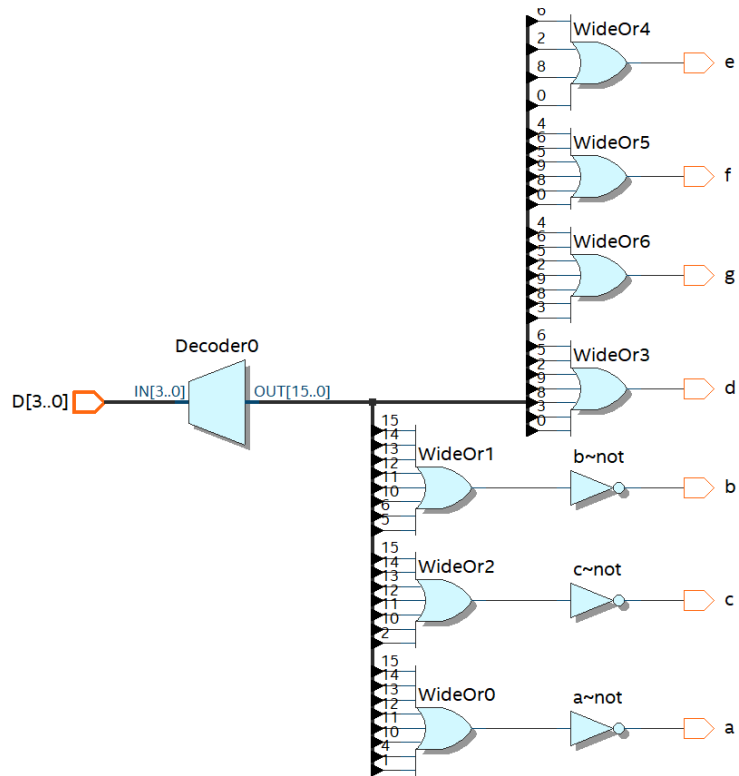


```
1 module tb;
2   reg[3:0]D;
3   wire a,b,c,d,e,f,g;
4   bcd_to_7segment dut(D,a,b,c,d,e,f,g);
5   initial
6   begin
7     D=4'b0000; #10
8     D=4'b1000; #10
9     D=4'b0111; #10
10    D=4'b1001; #10
11    D=4'b1100; #10
12    $finish;
13  end
14 endmodule
15
```

Waveform



Schematic



EDA Tools Used

- IntelQuartusPrime
- ModelSim

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